

2. General methods in studying constituents of crude drugs

2.1 EXTRACTION

Definition

- Extraction is the removal, withdrawal or separation of valuable substances from a biological matrix.
- is the process of obtaining the medicinally important components of a plant or a microorganism

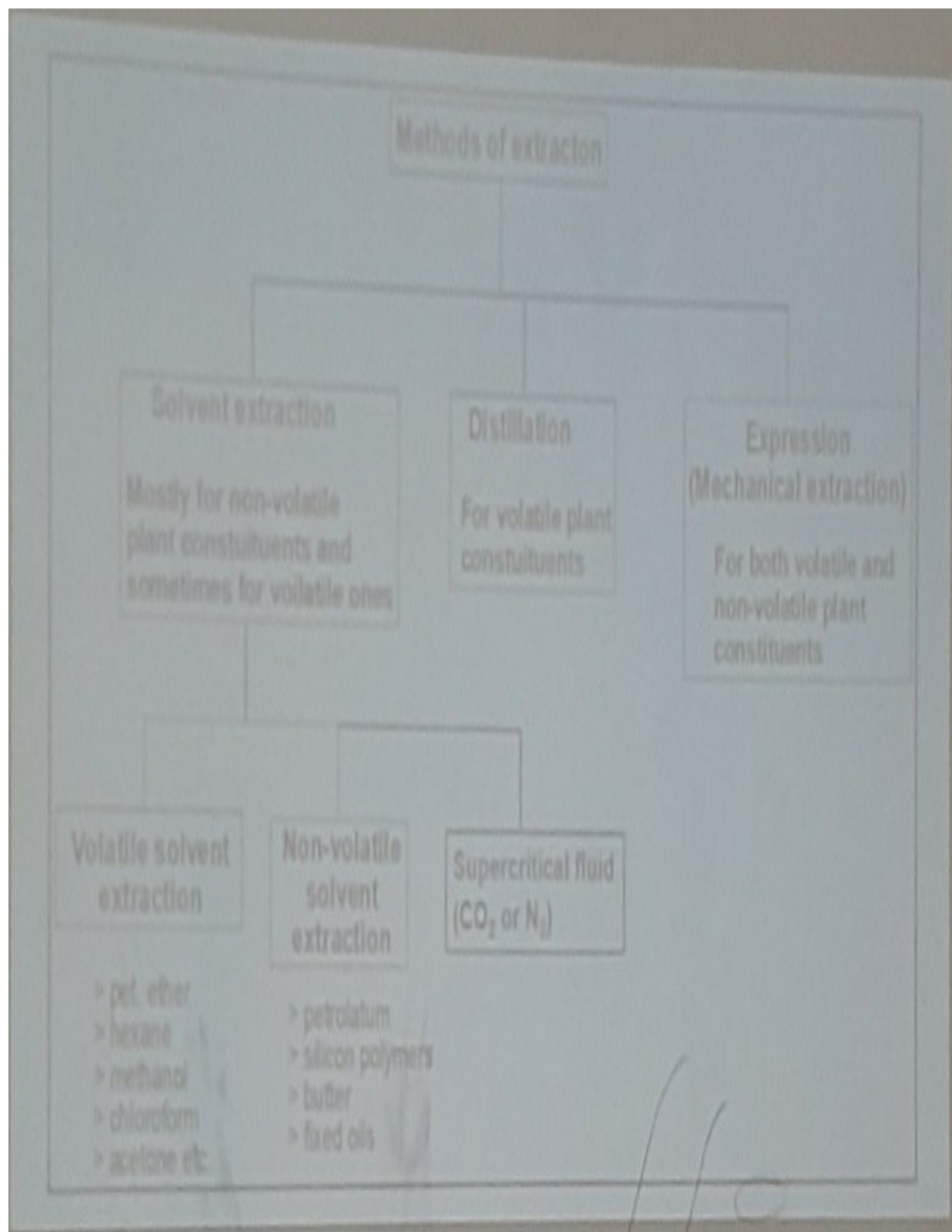


What is the need for extraction?

- To make palatable, concentrated preparations, eliminating the need to use bulky or distasteful drugs
- To create a product that contains only pharmacologically active constituents.
- To create a product of known strength and with reproducible activity
- To make the isolation of a pure active compound much easier

Ideal extraction method comprises:

- should be exhaustive
- Should preserve the plant constituents
- Preferably less time consuming
- Should be efficient
- Should be economical
- Should be selective



1. Solvent extraction

- Solvent extraction is a method of extraction where the plant material is extracted with a solvent, which could be liquid, semisolid or supercritical fluid (CO₂ or N₂).
- Mostly applied to obtain non-volatile plant constituents.
- Extraction is due to mainly solubility and to some extent due to diffusion
- Depending on the volatility of the solvent:
 - A. Volatile solvent extraction
 - B. Non-volatile solvent extraction
 - C. Supercritical solvent extraction

A. Volatile solvent extraction

Preparation of plant material for volatile solvent extraction

- Collection
- Garbling
- Drying
 - In most cases, the plant material is then dried but not always.
- Grinding
 - facilitates extraction by increasing surface area

Drying

- **purpose of drying:**

- Ease of transport, storage and ease of grinding
- Inhibit the growth of microorganisms
- Prevent enzymatic hydrolysis of active compounds

- **Two ways of drying:**

- **Natural drying**

- Under a shade (most of the time) and in sunlight

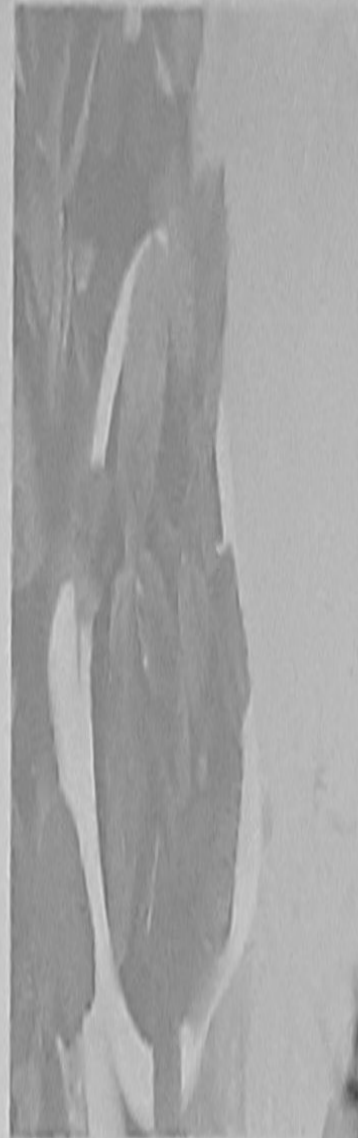
- **Artificial drying**

- Hot air drying, desiccant or by freeze-drying (sublimation)

General drying techniques of drying different plant parts

Leaves/Herbs:

- 40-50° C
- Maintains good color without overdrying herbs
- Normally destalked first
- Best dried for short periods of time only
prevents shrivelling and discolouration
- Leaves may be bleached through the drying
process best dried in dark



Root/ rhizomes

- Washed, Sliced and Dried between 30-65°C
- Temperature too high: forms a tough crust on the outside preventing the further drying of deeper tissue
- Thick organs may require long periods of drying completely
 - (10days-3weeks) to avoid mold growth.



Flowers

- Normally destalked before drying
- Flowers may also be bleached when dried
 - best dried in dark



Volatile oil containing herbs

- Air-Dried at very low temperature
- To avoid volatilization of oil contents
- Should also be dried separately from other herbs.
 - scent adheres to other herbs



Fruits and Seeds

- Normally partially dry before harvesting
- May be air-dried afterwards
- Seeds are normally separated from the fruits before drying
- Fruits are normally left whole
- Some fruits (cardamom) rupture due to excessive heat.



Bark

- Occasionally requires the removal of the outer layers
- Must be dried thoroughly to avoid microbial growth



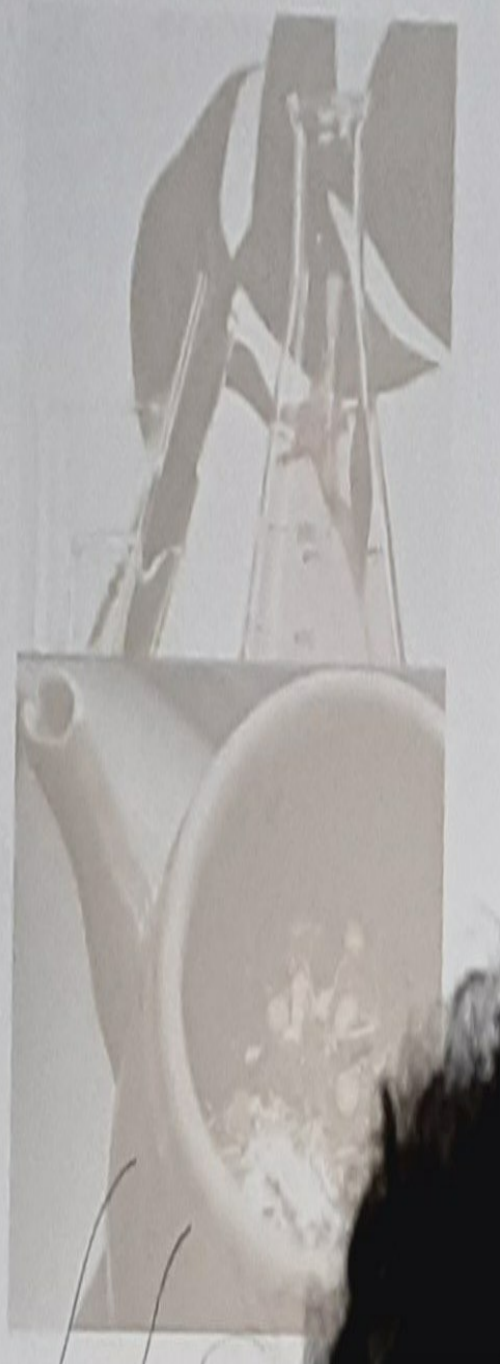
Extraction techniques

- **Two types of solvent extraction**

- Continuous: solvent recycled via a condenser or manually
- Batch: single contact between material & solvent

I. Infusion/ decoction

- **Infusion:** is extraction by hot solvent and the contact time between the plant material and the solvent is relatively short.
 - Good for fragile plant parts like leaves, buds or flowers.
 - Allowing the material to remain suspended in the solvent over time (a process often called steeping).



- An infusion is also the name for the resultant liquid.
- **Decoction:** plant material is boiled in the solvent.
 - Good for woody parts

- **Advantage:**

- Simplicity
- Quick method

- **Disadvantage:**

- Thermolabile compounds will be destroyed
- Wastage of solvent

II. Maceration/ percolation

- **Maceration:** is soaking the powdered plant material in any shaped glass or stainless steel container that is stoppered with occasional stirring.
 - Maceration is kept for 72hr and repeated accordingly until the solvent to be clear.
- **Percolation:** involves maceration followed by slow flow (slowly flushing with solvent) of the extract through the percolator.
- Usually done at room temperature.

Advantage:

- Simplicity
- Preserve thermolabile compounds

Disadvantage:

- Consumes too much solvent
- Saturation is attained quickly.
- Continuous stirring (shaking) might help.
- Time taking process with the possibly of hydrolysis of compounds

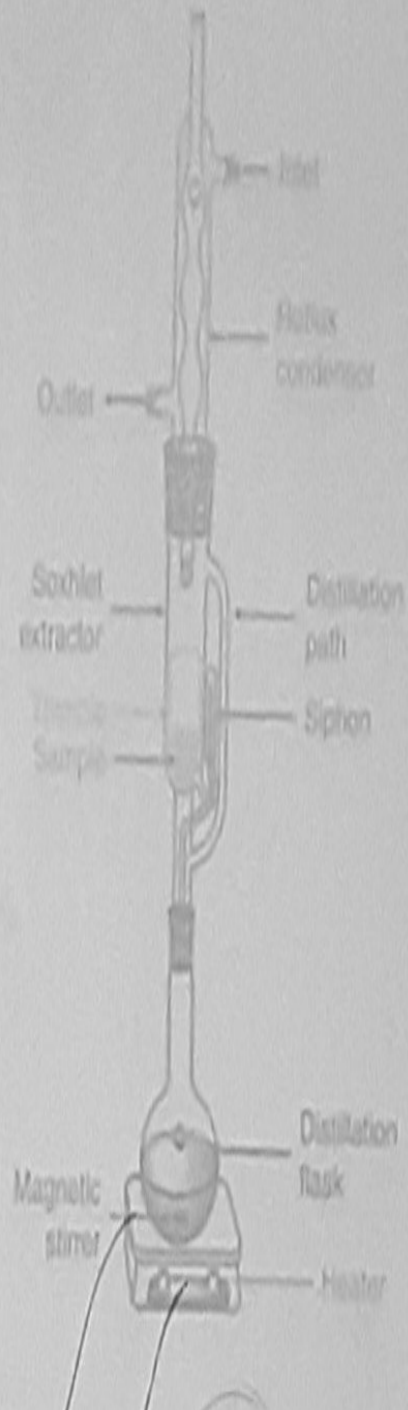


III. Soxhlet (continuous) extraction

- Components of soxhlet extraction

set up

- Heat source
- Flask containing solvent
- Extraction chamber
- Condenser connected to a flowing water
- Extraction timble (made of filter paper)



Advantage:

- Saves solvent
- Quick method
- Ease of changing solvent for the preparation of fractions

Disadvantage:

- Requires special apparatus
- Not applicable for thermolabile plant constituents
- No direct contact between plant material and heat source

IV Refluxing

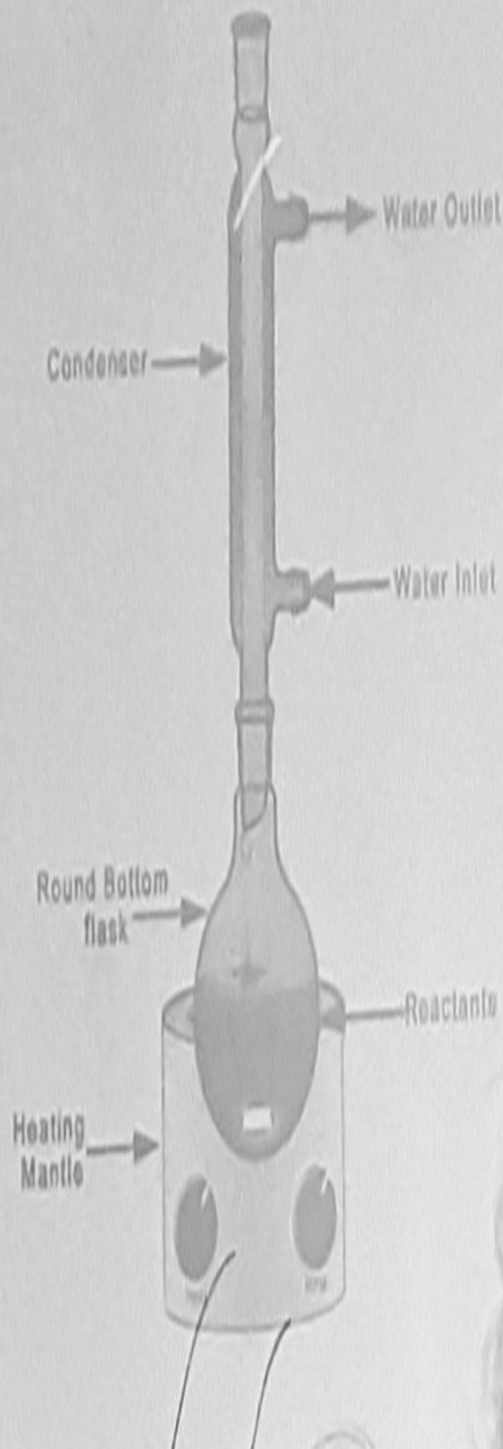
- Similar to decoction except that there is a condenser (continuous extraction)
- It is a constant volume extraction method because there is no wastage of solvent.

Advantage:

- Saves solvent
- Quick method

Disadvantage:

- Not applicable for thermolabile plant constituents

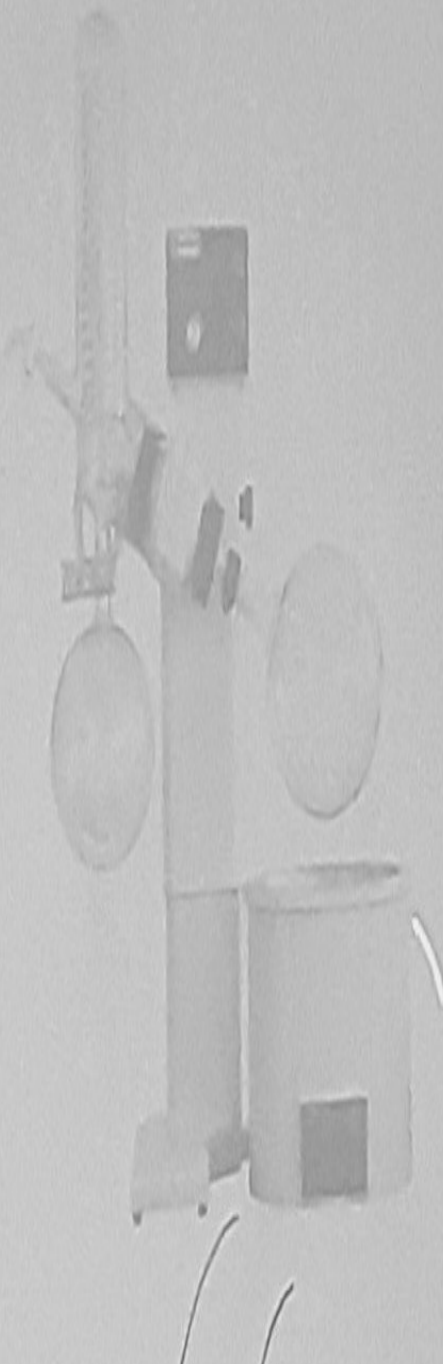


The ideal solvent for extraction should:

- Be highly selective towards the compounds to be extracted
- Inert towards intended compounds
- Affordable and accessible
- Be harmless to man and to the environment
- Be completely volatile or easily removed

Solvent Removal

- Rotary evaporation
 - At no more than 30 - 40 °C under negative pressure to preserve thermolabile constituents
 - Concentrated extracts should be stored in a refrigerator and a trace of toluene is added



B. Non-volatile Solvent Extraction

- The enfleurage method
 - Pressing between two plates
- The pneumatic method
 - Hot air flows between the material
- The maceration method
 - Macerate with a melted lard or oil



C. Supercritical Solvent Extraction

- Beyond its critical point, a fluid can have the density of a liquid and the viscosity of a gas
 - therefore it diffuses well through solids, and it is a good solvent
- Although, in theory, several gases can be used, the focus is exclusively on carbon dioxide
 - It is a natural product
 - Chemically inert
 - Non-flammable
 - Perfectly non-toxic,
 - Easy to eliminate completely
 - Selective, Readily available and Inexpensive